

HTML & CSS (Basics to Intermediate)

Conceptual:

1. What's the difference between `<div>` and `<section>`?
2. Explain the difference between inline, block, and inline-block elements.
3. What's the purpose of semantic HTML? Can you name a few semantic tags?
4. What are media queries, and how do you use them?
5. Explain the box model in CSS.
6. Difference between `position: absolute`, `relative`, `fixed`, and `sticky`.
7. How do flexbox and grid differ? When would you use each?
8. What's specificity in CSS and how does it work?

Practical:

- How would you center a `<div>` vertically and horizontally?
 - How can you make a responsive navbar without using a framework?
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JavaScript (Core + DOM + ES6)

Core Concepts:

1. Explain `var`, `let`, and `const`.
2. What are arrow functions, and how do they differ from regular functions?
3. What's the difference between `==` and `===`?
4. What are callbacks, promises, and `async/await`?
5. What's the event loop, and how does JavaScript handle asynchronous tasks?
6. Explain closures with an example.
7. What are higher-order functions?
8. Explain destructuring and the spread operator.

DOM & Browser:

1. How would you select elements using JavaScript?
2. Difference between `event.preventDefault()` and `event.stopPropagation()`.
3. How would you handle form validation using JS?

Practical:

- Implement a simple counter or todo list using vanilla JS.
- How would you fetch and display data from an API using `fetch()`?

Frontend Framework (React.js or similar)

1. What are components, props, and state in React?
2. What's the difference between class and functional components?
3. Explain the purpose of hooks (like useState and useEffect).
4. What is virtual DOM and why is it useful?
5. How does React handle re-rendering?
6. What are keys in React lists, and why are they important?
7. Have you used React Router or any state management library (like Redux or Context API)?

Practical:

- Describe how you'd build a simple blog app using React.
- How would you structure components for a dashboard?

Backend (for full-stack inclined candidates)

1. Have you used Node.js or Express? Explain the request–response cycle.
2. How would you handle routing and middleware in Express?
3. What's the difference between REST and GraphQL?
4. What database have you used (MongoDB/MySQL)? How do you connect it with your backend?
5. How do you handle authentication (JWT / sessions)?

Problem Solving & Thinking

1. You have a webpage that's loading slowly — how do you optimize it?
2. If your API call fails in production, how would you debug it?
3. How would you design a portfolio website so that it's maintainable and scalable?
4. What's one project you've worked on that you're proud of? What challenges did you face?

Behavioral / Soft Skills

1. What part of web development excites you the most — frontend, backend, or design?
2. How do you stay updated with new tools or frameworks?
3. How do you handle a bug that you can't solve for hours?

4. How comfortable are you working in a team or version-controlling with Git?